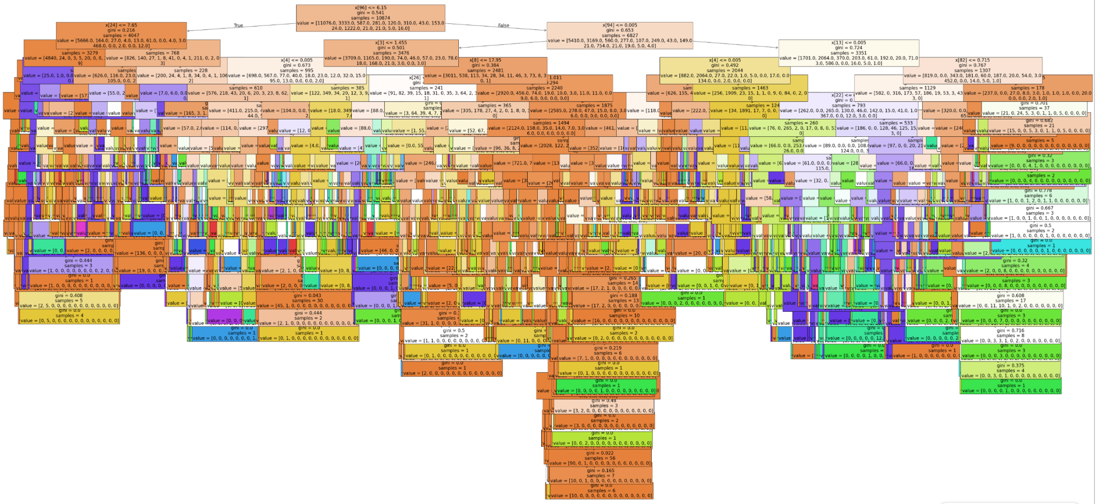
2.3 Answers

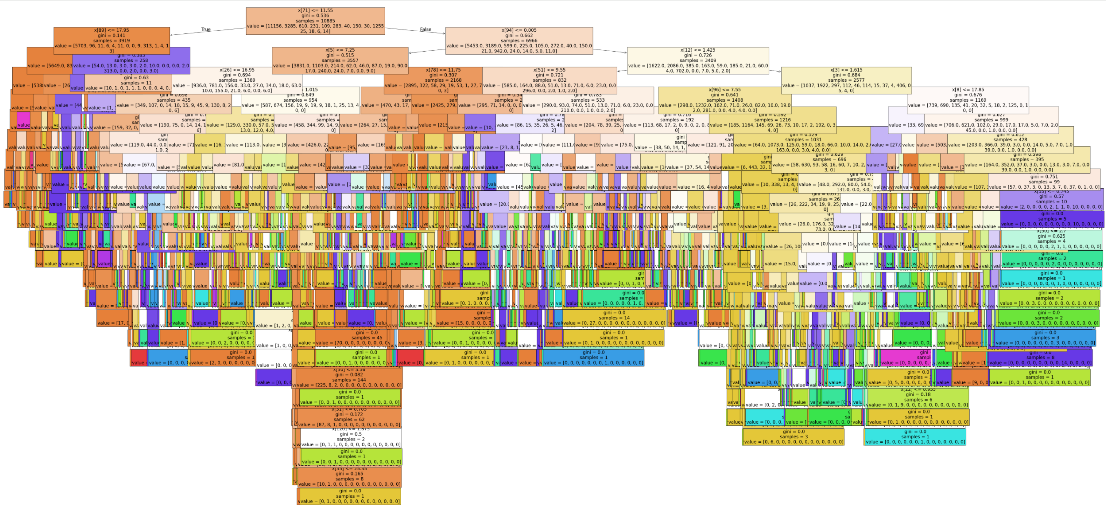
Estimator 15

```
fig = plt.figure(figsize=(80,40))
plot_tree(clf.estimators_[15], fontsize = 20, filled=True);
```

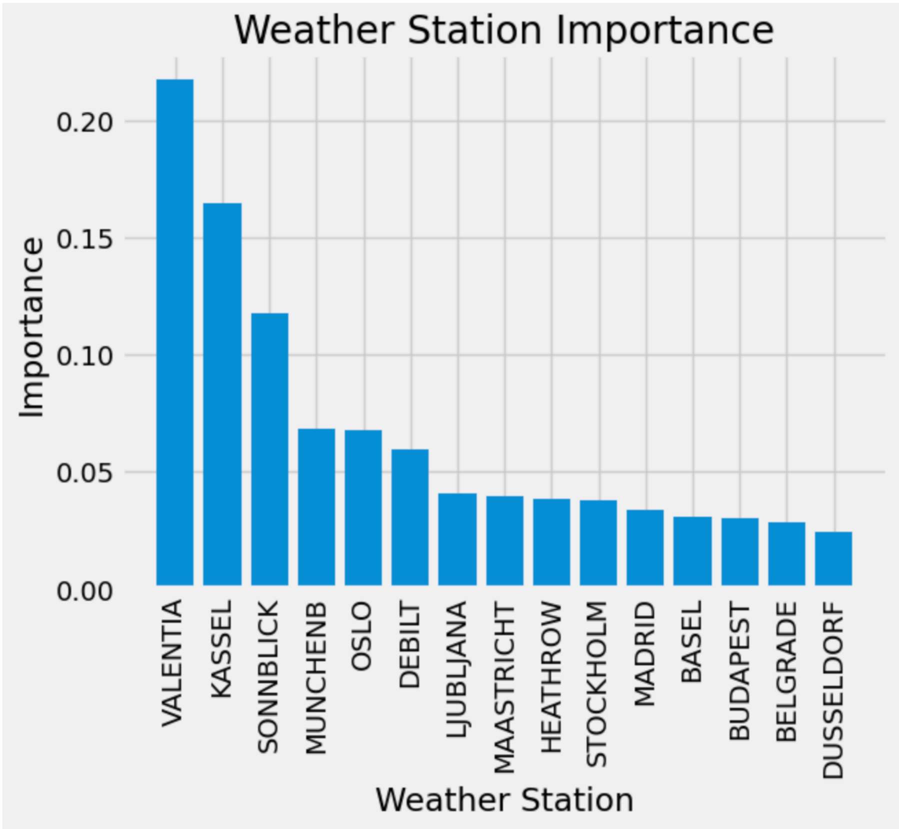


Estimator 35

```
fig = plt.figure(figsize=(80,40))
plot_tree(clf.estimators_[35], fontsize = 20, filled=True);
```



The test accuracy is 93%, training accuracy is 100%. However, the random forests are very difficult to understand which is to be expected when using the entire data set for all stations and years!



This shows Valentia, Kassel and Sonnblick are significantly the highest in terms of importance. This will be interesting to analyse further, as we already know there is an issue with Sonnblick's answer data.

VALENTIA

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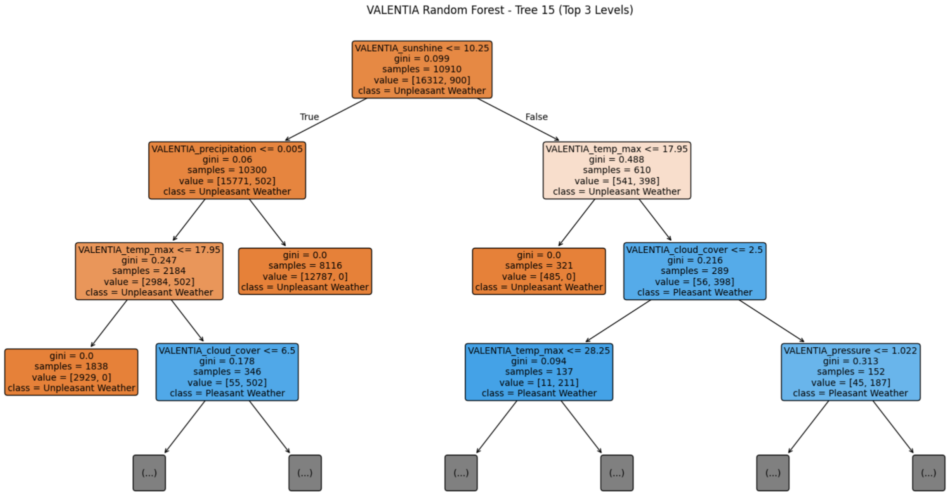
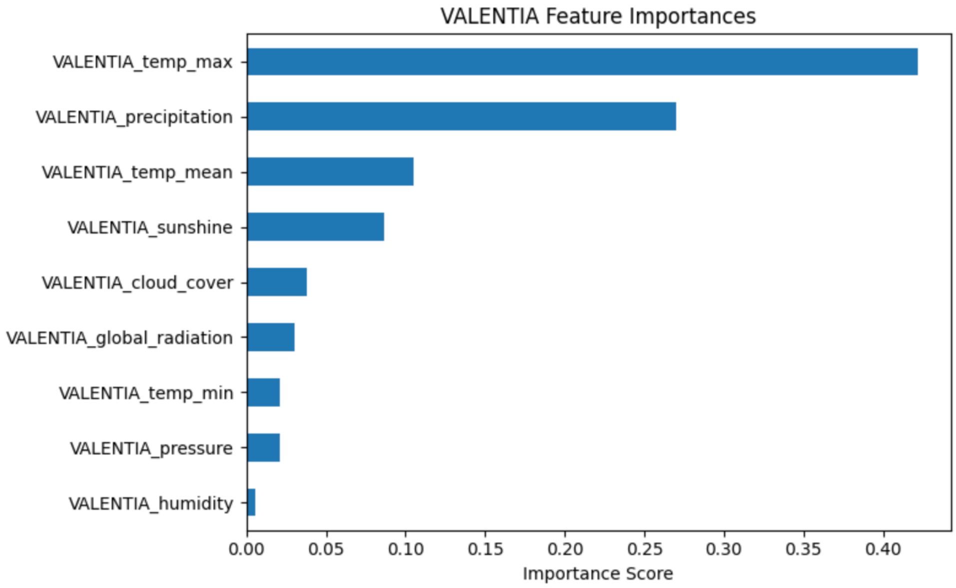


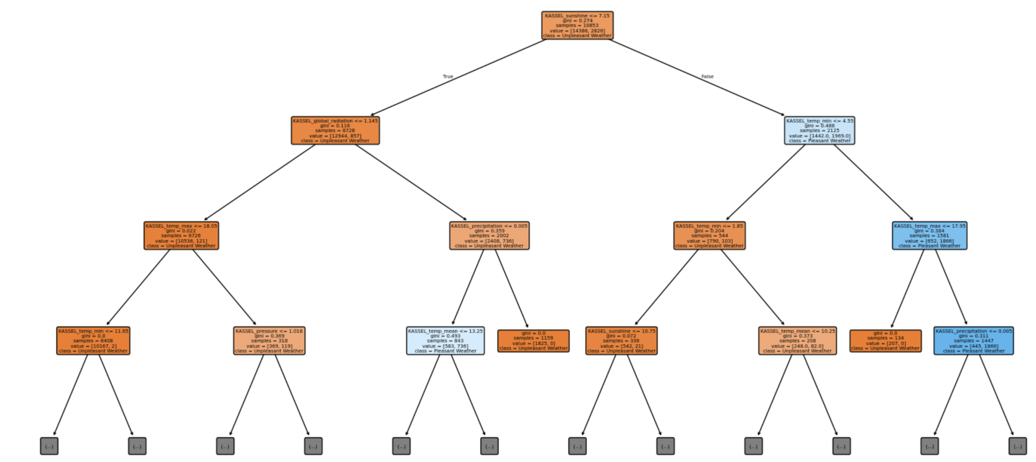
This makes sense as the first branch defines unpleasant weather when sunshine is under a certain score later followed by a lower max temperature, and then on the right defines pleasant weather when cloud cover is low and temperature max is not too hot!

Accuracy = 100% for both train and test sets.



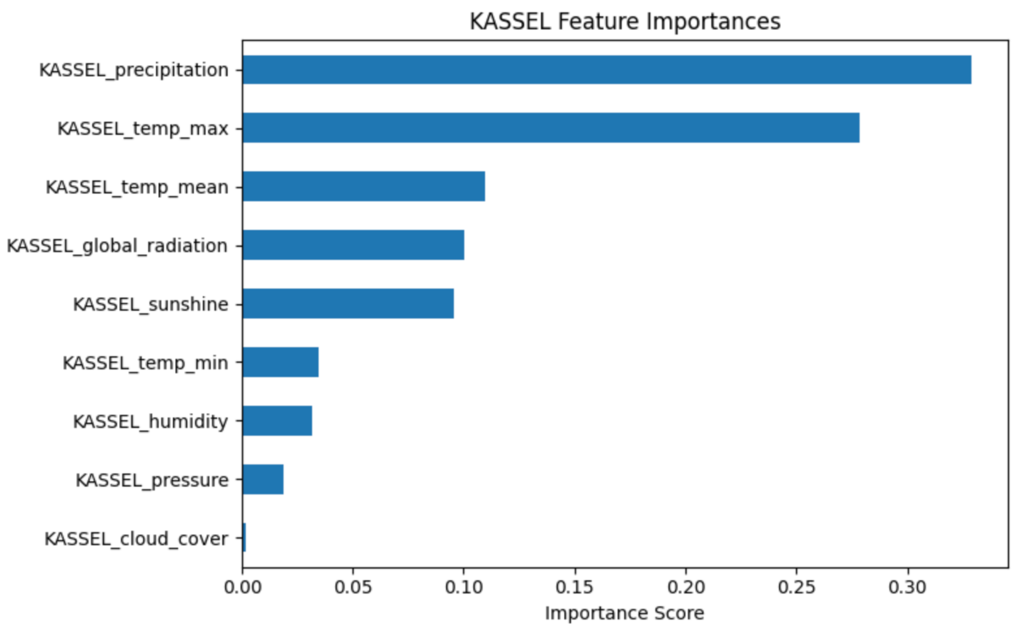
This also makes sense as we would expect a high temperature, rain and sunshine to be significant factors for what constitutes pleasant or unpleasant weather

KASSEL



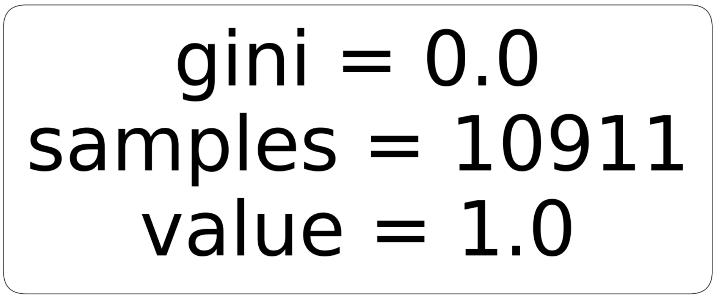
This makes sense as unpleasant weather defined by lower max and min temperatures low sunshine, and pleasant weather by lower rainfall.

Accuracy = 100% for both train and test sets

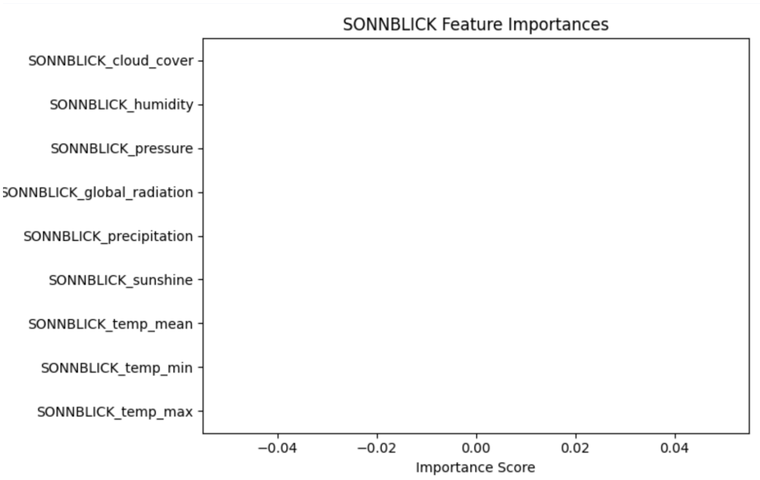


This is consistent with Valentia where rain and max and mean temperatures have the most importance in defining if a day has pleasant or unpleasant weather, with pressure and humidity similarly very low, although radiation has higher significance.

SONNBLICK



We already know the Sonnblick data is flawed so this is not surprising! This is still useful to show how data is portrayed when the original data is flawed.



The data clearly shows that max and mean temperature along with rainfall (precipitation) are the most important factors in determining whether a day is pleasant or unpleasant, closely followed by sunshine. I would recommend investing in climate-tracking equipment for these specific variables and to get the correct answer data for Sonnblick so we can further refine this analysis!